
Under-reported Harm: A Claim-level Audit of Abstract Reporting on AI Assistance and Human Learning

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Abstract

AI assistance is rapidly entering clinical training and practice, and evidence of agents’ benefits and harms to human learning is required to develop evidence-based guidelines for governance in academic medical centers. To assess how current evidence is being reported, this study extracted 6,063 atomic claims from the abstracts of 1,402 papers (2015–2026) and coded each into a 19-field predicate–argument schema. Within the same paper, harm claims are hedged at about twice the odds of benefit claims (OR 2.02, 95% CI 1.49–2.72) and marked statistically significant at about one-seventh the odds (OR 0.14, 0.08–0.27), and both differences hold within generative-AI studies. Across the corpus, 82.1% of harm claims report no statistical test in the abstract, against 70.8% of benefit claims. The share of harm claims is inversely correlated with quantitative reporting (Spearman $\rho = -0.46$, 95% CI -0.59 to -0.11 , across research areas): among claims coded beneficial or harmful, the harm share is 39.8% for behavioral-strategy outcomes (8.4% of the corpus) and 37.4% under longitudinal measurement (2.9% of the corpus). These results indicate that abstract-level under-reporting of harms is pervasive in studies of AI assistance in learning.

1 Introduction

AI assistance, increasingly in agentic form, is entering clinical work faster than the evidence needed to govern it. AI-assisted study guides now support medical education [1], AI systems support diagnostic reasoning [9] and scribing [11], and coding and billing are increasingly automated. Procedural coding already presumes that these uses can be sorted by degree of machine involvement: assistive, where the machine detects clinically relevant data without analyzing it; augmentative, where it analyzes or quantifies data to yield a clinically meaningful output; and autonomous, where it interprets data and generates conclusions without concurrent physician involvement [10].

While the procedural taxonomy classifies the system’s role in a clinical workflow, it does not describe the system’s role relative to that of the clinician or the trainee. Deciding when AI assistance should be offered — at which workflow stage, to a learner at which point in training — requires knowing which form of involvement affects which learning target, for whom, by what mechanism, and over what interval [36].

To support efforts to develop an evidence-based framework for calibrated AI integration in clinical workflows, we asked a prerequisite question: can the existing literature support the type of evidence-based inference that healthcare requires? Treating the literature itself as the object of measurement, this work contributes a claim-level semantic audit of how the evidence on AI assistance and human

learning is reported in abstracts, covering 6,063 coded claims from 1,402 papers. The corpus covers machine assistance broadly, as the cognitive effects at issue are not specific to one technology; clinical and agentic deployment motivates the audit, and we report generative-AI and clinical subsets separately. Four questions organize the analysis herein:

1. *Does the corpus vary primarily by the content of claims, or by how completely they are specified?*
2. *Do benefit and harm claims differ in evidential form, and is any difference compositional?* We expected an asymmetry in framing, hedging and statistical marking, and tested whether it survives adjustment for claim specificity, publication format, and comparison within the same paper.
3. *What do abstracts report about testing harm?* An abstract can report a significant test, a null test, or no test at all; it cannot show whether an outcome reported without a test was never tested or was tested and left out. We therefore separate reported null tests from absent reports, and relate harm rates to how often each outcome type, setting and horizon is reported at all. Whether harms were measured in the underlying studies requires the full texts (Section 4).
4. *What happens to the aggregate picture when claim specificity is equalized?* Claims differ in how much they commit to, from a category through a direction to a size. If quantified claims are unevenly distributed across outcomes, a filter that favors quantified evidence — a common standard in guideline development — will bias the conclusions derived. We standardized the corpus to equal shares of taxonomic, directional, and magnitude claims.

2 Methods

2.1 Evidence search

We applied a population-concept-context frame, as specified by Peters et al. [12]. A source is eligible if it concerns human knowledge retention, skill maintenance or cognitive performance under machine assistance, automation or generative-AI support. This wording excludes an adjacent literature on offloading to physical artifacts such as notes, diaries and placed objects.

- Included: human–automation interaction and human factors; cognitive psychology of knowledge types, retrieval and metacognition; empirical studies of generative-AI assistance effects. Original research, meta-analyses and systematic reviews; professional-body standards admitted as evidence of institutional uptake rather than of effect; preprints admitted and labeled.
- Excluded by scope decision: medical-education assessment frameworks; tacit-knowledge and apprenticeship literatures; opinion and perspective pieces.
- Window: 2015 onward.

Pre-window works were admitted under a documented exception rule: a source predating the window is eligible if (i) it is cited as the origin of a construct the framework uses by two or more included in-window sources, and (ii) no in-window substitute establishes the same claim.

Records were gathered through two channels. Channel 1 searched OpenAlex (1,396 records), PubMed (457) and arXiv (427) with preregistered query families. Channel 2 performed inclusion-gated citation chasing: backward and forward chasing from the 279 included channel-1 records, seeded only by included works, with per-seed caps of 100 forward and 100 backward citations declared in configuration. 267 of the 279 included channel-1 records resolved to an OpenAlex identifier and served as seeds; the other 12 could not be chased. Chasing to depth 1 contributed 10,226 records; a depth-2 generation was chased and cached but its screening was declined on cost grounds, so no depth-2 record enters the corpus. A third channel (author-supplied seed queries) was declared in the protocol and not executed; it is reported as not executed rather than as yielding zero. The 12,506 records identified were deduplicated on DOI and then on a normalized title-and-year key, leaving 11,367 (Figure 1).

Reporting was patterned after PRISMA-ScR [13], with search reporting per PRISMA-S [14] and automation-tool disclosure per PRISMA 2020 item 27 [15].

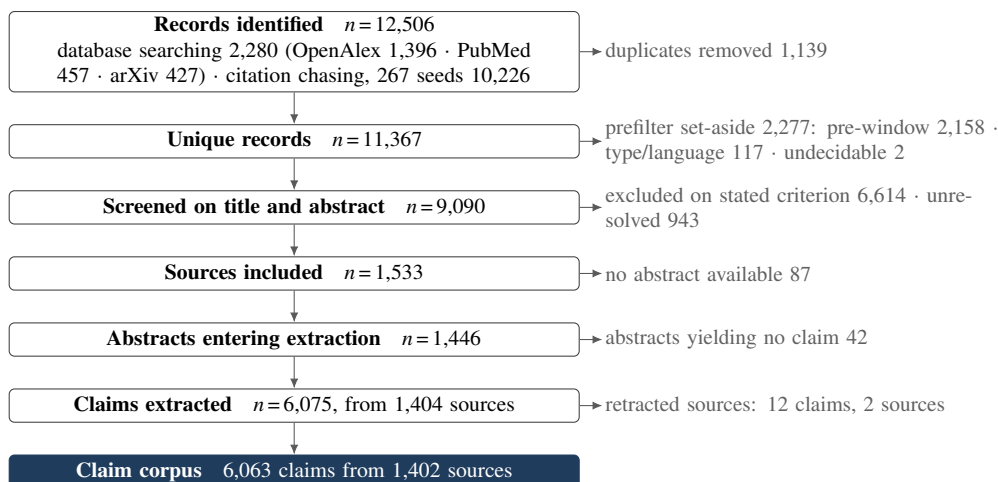


Figure 1: PRISMA-ScR flow from record identification through claim extraction. Boxes on the left carry the records or claims retained at each stage; boxes on the right record what left the pipeline and why. The corpus is abstract-derived throughout: no full-text eligibility stage was performed.

2.2 Screening

A deterministic prefilter was run on publication year, language and publication type. Language matching accepts both ISO 639-1 and 639-2 codes, PubMed emitting the latter. Of the 11,367 deduplicated records, 9,090 passed, 117 were excluded by rule, 2 were undecidable, and 2,158 published before the 2015 window were deferred — not excluded — pending the pre-window exception rule.

The 9,090 passing records were screened on title and abstract by gpt-5.6-luna under a three-stage cascade governed by one asymmetry: a title alone may exclude a record but may never include one.

1. Stage 1 screens title and abstract together and returns include, exclude or unclear with a stated failed criterion; an exclusion on grounds of insufficient information is coerced to unclear.
2. Stage 2 recovers missing abstracts from Semantic Scholar and re-screens.
3. Stage 3 screens the remaining abstract-less records on title only and may only exclude or defer.

Screening yielded 1,533 included records, 6,614 excluded and 943 unresolved. Included sources were checked for retraction against three signals: the Crossref update-to relation specified in the protocol, the OpenAlex `is_retracted` flag, and a marker scan of deposited titles. See Appendix A.1 for details of the pre-window exceptions.

2.3 Retrieval

Full-text retrieval was piloted through arXiv, OpenAlex open-access links and Europe PMC on an earlier 50-source sample; most sources resolved as abstract-only. The claims pipeline therefore operates on abstracts by design, and retrieval depth is carried on every extracted claim so that the abstract-only basis of the corpus remains explicit (see Limitations).

2.4 Claim extraction and validation

Claims were extracted from the abstracts of 1,446 records (of 1,533 total). The extractor used was claude-haiku-4-5, temperature 0, seed fixed, output forced through a constrained tool schema. Each claim comprises a normalized proposition; a role, signature (what effect was observed) or mechanism (why or how); a claim specificity; and a verbatim supporting quote from the abstract. Claim specificity records how much a claim commits to: *taxonomic*, a category or definition is coherent and usable;

directional, an effect exists and runs in a stated direction; or *magnitude*, an effect has a stated size or holds under a stated moderator. It is assigned by a priority-ordered rule — magnitude before directional before taxonomic — because a claim that reports a number also states a direction, and without an order of application the label is arbitrary. Claim specificity describes the claim; the corpus has no reliable study-design variable (Section 3.3). Extraction yielded 6,075 claims from 1,404 sources (42 abstract-bearing sources yielded none); integrity exclusion leaves 6,063 claims from 1,402 sources, between 1 and 16 per source, mean 4.32. Every claim carries `retrieval_depth` = “abstract only”: abstracts state findings rather than pathways, so mechanism claims (1,631, against 4,432 signature) under-represent whatever mechanisms appear only in body text.

2.5 Claim schema

A predicate-argument schema was chosen for parsing claims [16–18]. A subject–relation–object representation with attached modifiers is the common structure for claims extracted from biomedical and behavioral abstracts. SemRep’s predications [2], the SciClaim graph schema [16], the five-slot “super-pattern” derived from cross-disciplinary claims [17], and schema-based slot-filling of clinical-trial abstracts [18] converge on a common set of features, including factuality, statistical marking, direction of effect, and population.

Treating the strength of the subject–object link as an ordered categorical variable is established at corpus scale. Yu et al. classified roughly 38,000 abstracts into no relationship, correlational, conditional causal and direct causal [3], and Haber et al. rated 1,170 abstracts on a none/weak/moderate/strong scale, finding that 45.7% of linking statements used “associate” as their root and that reviewers disagreed substantially about the causal force of individual words [4]. Our `relation` and `epistemic_strength` fields are close to a merge of those two scales.

To inform the parsing schema, 156 claims were randomly selected and stratified by role and claim specificity. Sample claims were parsed according to the schema by two LLMs (Claude Haiku 4.5 and OpenAI Luna), and the agreement between them was calculated as per-field Cohen’s κ , with source-blocked bootstrap CIs, and a pre-established floor of 0.60.

After parsing, free-text entity labels were canonicalized (case, punctuation and synonym variants mapped to one surface form); the categorical fields were not altered.

The claim-specificity rule was validated by inter-model agreement on the 241-claim extraction that preceded the corpus-wide pass. An initial version using unordered category glosses measured Cohen’s κ of -0.207 against an independent model — below chance — with zero agreement on the taxonomic category; the conclusion that had been drawn from those labels was retracted, the glosses were replaced by the ordered rule with worked contrasts, and κ rose to 0.634. Holding the validator’s prompt at its original wording reproduces 0.634, so the gain is attributable to the extractor’s definitions rather than to alignment of the two models’ codebooks.

Recording the kind of evidence each assertion offers follows biocuration practice, which also records whether a human or a machine made each annotation [20, 21]. We stop short of scoring evidence, as PharmGKB does [22]: formal grading is unreliable even in expert hands, with GRADE and CINeMA disagreeing on 34% of 105 comparisons in one network meta-analysis [25], and automated grading performs worse [23, 24]. Thus, the taxonomic/directional/magnitude scheme is descriptive, with its fields as unweighted nominal indicators.

Finally, every claim was parsed into a 19-field predicate–argument frame by claude-haiku-4-5 under a constrained schema (see Table 1). Absence is explicit throughout: `not_stated` (the claim is silent) and `not_applicable` (the slot cannot apply to the chosen relation) are distinct values. Frames were produced through the synchronous API and the Anthropic Message Batches API. The frame set joins the claim table one-to-one, asserted in the notebook, with 96.2% of evidence spans verbatim substrings of their source quote. Results for reliability of coding can be found in Appendix A.2.

2.6 Exploratory claim analysis

Multiple Correspondence Analysis (MCA) and Latent Class Analysis (LCA) were used to explore the semantic structure of claims. Correspondence analysis of a cases-by-codes matrix has been proposed as an alternative to narrative cross-case synthesis [26], and iterative MCA has been demonstrated

Table 1: The claim schema consists of 19 fields.

Field	Type	Values
<i>subject_entity</i>	free text	the intervention as worded
<i>subject_type</i>	nominal	generative AI assistant · decision support · search/retrieval · automation · calculator/tool · non-AI aid · none (observational)
<i>subject_mode</i>	nominal	availability · use · reliance · removal
<i>object_entity</i>	free text	the affected construct as worded
<i>object_type</i>	nominal	task performance · cognitive process · disposition · subjective experience · physiological · behavioral strategy
<i>relation</i>	nominal, ordered	mediates > moderates > causes > associates > no_relation
<i>mediator</i>	free text	intervening variable
<i>moderator</i>	free text	conditioning variable
<i>direction</i>	nominal	increase · decrease · no_change · non_monotonic
<i>valence</i>	nominal	beneficial · harmful · neutral · contested
<i>magnitude_value, magnitude_unit</i>	numeric	stated effect size only
<i>statistical_marking</i>	ordinal	significant · not_significant · not_stated
<i>epistemic_strength</i>	ordinal	asserted · hedged · speculative
<i>population_role</i>	nominal	students · clinicians/professionals · workers · general adults · children · mixed
<i>population_setting</i>	nominal	education · clinical · workplace · laboratory · everyday
<i>measurement_horizon</i>	nominal	immediate · delayed · longitudinal
<i>n_reported</i>	numeric	sample size if stated
<i>evidence_span</i>	verbatim	substring of the quote carrying the relation

as a route from qualitative codes to dimensional structure [27]; interpretation by category and field contribution follows accepted practice [28].

3 Results

3.1 Evidence coverage

The corpus comprises 6,063 semantically coded claims drawn from 1,402 papers published between 2015 and 2026, with 62% of claims coming from 2025 and 2026. Papers contribute between 1 and 16 claims each (mean 4.32, median 4), so all subsequent inferential models cluster standard errors on the paper and all coverage counts use unique papers as the denominator.

Every claim was extracted from an abstract; the corpus contains no full-text retrievals. We emphasize that the value `not_stated` records what an abstract — not a full-text study — omitted. Field-level stated rates vary by more than a factor of six, from 99.5% for `epistemic_strength` down to 19.9% for `statistical_marking` and 15.9% for `measurement_horizon`.

Extraction used three model configurations across three retrieval channels. To establish that downstream distributional findings are not artifacts of the pipeline, we computed the total variation distance between each configuration’s field distribution and the pooled distribution. The maximum deviation across all 19 fields is 0.06 for any single field and 0.11 for the composite relation–direction–valence signature.

3.2 Claim schema

The claim schema consists of 19 fields (see Table 1). Only 7 fields achieved $\kappa \geq 0.60$ on the first pass. Upon examination of the discrepant categories, “mediator,” “moderator,” and “measurement_horizon” were found to be sparsely populated. This reflects a frequent finding in the LLM-as-judge literature [19].

3.2.1 Exploratory analyses

Clustering the 19 fields on pairwise adjusted mutual information (AMI), which corrects for chance, recovers a quantitative block (magnitude, unit, sample size), a population block, an object block, a large assertion block, and three fields that join nothing (`mediator`, `moderator`, `evidence_span`); the associations are driven by co-absence (Appendix B.4).

Table 2: Odds ratios [95% CI] for harmful versus beneficial claims under four adjustments. M0–M2: logistic regression on all 3,879 beneficial-or-harmful claims from 1,286 papers, with paper-clustered standard errors. M3: conditional logit comparing claims within the same paper, estimable only from papers with within-paper variation in both valence and the outcome; the last column gives the papers and claims that contribute.

Outcome <i>adjustment</i>	M0 <i>none</i>	M1 +specificity	M2 +format	M3 <i>within paper</i>	M3 <i>papers / claims</i>
Associative framing [†]	2.27 [1.75, 2.94]	2.27 [1.75, 2.94]	2.27 [1.75, 2.94]	1.61 [1.11, 2.33]	116 / 479
Hedged or speculative	2.44 [1.99, 3.01]	2.17 [1.75, 2.69]	2.17 [1.75, 2.71]	2.02 [1.49, 2.72]	178 / 734
Significance-marked	0.21 [0.15, 0.29]	0.23 [0.16, 0.33]	0.23 [0.16, 0.34]	0.14 [0.08, 0.27]	81 / 377
Carries a magnitude	0.43 [0.27, 0.68]	1.00 [0.54, 1.86]	0.98 [0.54, 1.77]	0.67 [0.36, 1.26]	39 / 174

[†]Depends on `relation`, whose inter-model agreement (κ 0.532) is below the 0.60 floor; provisional.

Multiple Correspondence Analysis (MCA) also partitioned the data on dimensions related to missingness, valence, and null findings (details in Appendix B.2). LCA revealed that the data supports approximately 9–12 classes, and we report $K = 11$ (entropy $R^2 = 0.93$) (see Appendix B.3).

3.3 Benefit and harm claims differ in evidential form

Restricting to the 3,879 claims coded beneficial or harmful (1,286 papers), we regressed four indicators of evidential form on claim valence. Claim specificity and publication format are candidate confounders, since harm rates differ by specificity (9.2% of magnitude claims, 18.9% of directional and 25.3% of taxonomic) and by format (18.1% of articles, 27.6% of preprints).

Adjustment on `work_type` is not available: the field carries no study design information for 5,306 of 6,063 claims (1,236 of 1,402 papers), as Crossref emits a bare format label while only PubMed-indexed records carry design terms. The entire corpus contains 11 papers tagged as randomized trials, 4 as systematic reviews and 3 as meta-analyses. Thus, we adjust on claim specificity, which is complete; on publication format; and — most stringently — by comparing only claims from the same paper, using a conditional logit that holds fixed everything about the paper, including publication type, venue, year, topic and authorship. Three hundred and twelve papers contain both a beneficial and a harmful claim. Each fixed-effects estimate uses only those that also vary on the outcome in question (Table 2), so it describes papers that report both kinds of claim; these are somewhat more recent, more often about generative AI, and longer in claims per abstract than the rest (Table A4).

Hedging and significance marking differ under every adjustment, including the within-paper comparison (Table 2). Associative framing also differs, but it depends on `relation`, which was the only field below the reliability floor, and loses significance within generative-AI studies; we did not rely on this result. The magnitude difference disappears once claim specificity enters the model, as expected.

Subsets and sensitivity. Within generative-AI claims (2,408 claims, 873 papers) the within-paper estimates are 2.31 [1.57, 3.42] for hedging and 0.08 [0.03, 0.20] for significance marking (Table A3). Clinical claims (352 claims, 193 papers) are too few for the within-paper model, since only nine papers report both a benefit and a harm, but the pooled estimates are directionally consistent (hedging 3.11 [1.70, 5.68]; significance marking 0.20 [0.09, 0.47]). The estimates are also directionally consistent within each retrieval route (Table A6). To test whether this asymmetry might be attributable to coding error, we randomly flipped 5–20% of valence labels, which pulled every estimate toward 1 (Table A5).

In the abstracts, most claims report no test: 82.1% of harm claims carry `statistical_marking = not_stated`, against 70.8% of benefit claims, and a reported null test is rarer still (`not_significant`, 186 claims across all valences). An abstract cannot show whether an outcome reported without a test was tested in the study.

The share of harm claims is inversely correlated with quantitative reporting. Across the 25 outcome $\times$ horizon $\times$ setting cells holding at least 20 beneficial-or-harmful claims, a cell’s harm share and the share of its benefit claims that report a statistical test correlate at Spearman $\rho = -0.46$ (95% CI

Table 3: Field distributions among claims stating the field, raw and standardized to an equal taxonomic/directional/magnitude mix.

Level	Raw %	Balanced %	Shift (pp)
valence = beneficial	62.2	67.1	+4.9
valence = harmful	14.8	12.4	-2.3
relation = causes	59.4	64.1	+4.7
relation = no relation	5.6	3.8	-1.8
direction = increase	71.1	75.2	+4.1
object_type = task performance	33.4	41.8	+8.4
object_type = cognitive process	30.0	26.7	-3.4
statistical_marking = significant	84.6	92.8	+8.2
measurement_horizon = longitudinal	18.3	12.5	-5.8

-0.59 to -0.11, bootstrapped over papers). Benefit claims are used so that harm claims’ own lower testing rate cannot drive the correlation. Taking a category’s harm rate as the share of harmful claims among its claims coded beneficial or harmful, harm makes up 39.8% of behavioral-strategy claims, although that outcome type holds only 8.4% of the corpus, and 37.4% of claims under longitudinal measurement (2.9% of the corpus). Behavioral strategies are often studied in real-world settings, where controlled measurement is harder, which may limit how quantitatively such claims are reported. Balancing the margins moves the overall harm share in both directions. Among claims stating each field, balancing measurement horizon raises it from 19.2% to 23.2%, outcome type from 18.2% to 21.6%, and population setting from 14.5% to 25.3%, while balancing claim specificity lowers it from 19.2% to 17.8%.

3.4 Standardizing claim specificity

To ask what the literature would look like if quantified claims were as common as directional and taxonomic ones, we reweighted claims by direct standardization. Each paper first receives unit weight split across its claims, removing the influence of verbose abstracts; the observed split of 60% directional, 10% magnitude and 30% taxonomic claims is then rebalanced to equal thirds. Paper weighting alone was found to move every quantity by less than 1.5 percentage points, so the shifts in Table 3 reflect the specificity rebalancing.

Giving magnitude claims equal weight yields a more favorable portrait of AI assistance. Beneficial claims gain 4.9 points, causal framing gains 4.7, and significance-marked findings gain 8.2. The balanced corpus is also more concentrated on task performance (+8.4 points) and less on cognitive process (-3.4), and it loses roughly a third of its longitudinal share, as magnitude claims in this literature are disproportionately short-horizon.

3.5 Coverage gaps

Marginal counts overstate what the literature covers, as fields are only jointly observable on a fraction of claims. Only 2,102 claims (34.7%, 981 papers) state intervention type, outcome type and setting together, and only 530 claims (8.7%, 344 papers) state outcome, population and measurement horizon together. We call a cell under-represented when its observed count falls well below the count expected if its margins were independent. The baseline is descriptive: a low cell may be a combination that is rare for good reason, and every count describes the retrieved corpus rather than the literature as a whole.

4 Discussion

Our results add to a substantial body of evidence on the under-reporting of harms. Ioannidis and Lau found adverse-effect severity adequately defined in 39% of trial reports, a median of 0.3 page given to safety results, and less space given to safety in trials that reported significant efficacy [29]. CONSORT Harms 2022 modified thirteen checklist items and added three more because the 2004 harms extension had not been consistently applied [30]. Dwan et al. measured selective reporting directly, comparing protocols against publications across 3,140 trials [31]. These studies looked expressly for studies that measured harm and then reported the results poorly.

Table 4: Under-represented cells in the intervention $\times$ outcome $\times$ setting cube, ranked by shortfall against the independence baseline. GenAI = generative AI assistant.

Cell	Claims	Papers	Expected	Ratio
GenAI $\times$ task performance $\times$ laboratory	9	9	32.2	0.28
GenAI $\times$ cognitive process $\times$ laboratory	1	1	20.2	0.05
GenAI $\times$ subjective experience $\times$ clinical	10	9	27.5	0.36
GenAI $\times$ disposition $\times$ clinical	8	7	23.7	0.34
GenAI $\times$ disposition $\times$ laboratory	1	1	8.3	0.12
<i>Non-AI aid</i> $\times$ task performance $\times$ laboratory	31	25	7.8	3.97

Our analysis asks a different question of the published record: whether harm and benefit claims made in the same abstract carry the same evidential form. First, our results show that the asymmetry is not an artifact of where harm claims come from. Hedging (OR 2.02) and significance marking (OR 0.14) survive conditioning on claim specificity and publication format, survive a comparison of claims within the same paper (Table 2), and hold within generative-AI studies. Harm claims about AI assistance and learning carried fewer statistical tests in their abstracts: 82.1% report no statistical test, compared to 70.8% of benefit claims.

Second, the analysis revealed that harm registers either as a decline in commonly used constructs, which accounts for 496 claims spanning critical thinking, learning outcomes, spatial knowledge acquisition and originality, or as a rise in poorly measured constructs, which accounts for 190 claims spanning cognitive load, automation bias, over-reliance, deskilling, dependency and the forgetting of offloaded material. A 2026 scoping review finds evidence of clinical deskilling “scarce” but consistent across specialties [33]. This audit suggests one reason that evidence appears scarce: claims about constructs like deskilling are mostly coded as behavioral strategies, an outcome type that holds 8.4% of the corpus and in which 39.8% of beneficial-or-harmful claims are harms.

Third, the within-paper asymmetry in Table 2 is best contextualized within the meta-research literature on harms reporting. That literature has established that adverse outcomes are systematically under-reported relative to benefits, chiefly by comparing published trial reports against their unpublished counterparts: Golder et al., synthesizing 28 methodological evaluations, found that adverse-event information present in unpublished study documents reached the published record less than half the time [35]. Those designs require an unpublished comparator and operate at the level of the study. Our contribution shows that the same asymmetry is detectable in the published abstract alone, at the level of the individual claim: within a single abstract, the same authors hedge their harm claims at twice the odds of their benefit claims (OR 2.02) and mark them statistically significant at one-seventh the odds (OR 0.14). Thus, claim-level extraction recovers a signature of selective harms reporting without access to protocols, unpublished data, or a paper’s full text, which is useful in fast-moving areas dominated by non-trial evidence.

Fourth, the content omitted by claims varies more than what they include. Generative AI is under-represented in laboratory settings in the retrieved corpus; our analysis only uncovered one claim, from one paper, on generative AI and cognitive process under laboratory conditions, against 20.2 expected under independence, while controlled laboratory work on calculators, notes and search is over-represented (Table 4).

For governance in academic medical centers, the results show that harm evidence is less often and less completely reported in abstracts; we emphasize that these results do not show that harms went unmeasured, nor do they estimate how often AI assistance causes harm. Nonetheless, governance frameworks that rely on reported evidence risk inheriting the reporting gap, and infrastructure that enables titrating the dose of AI-augmented decision support for the stage of the learner becomes harder to justify. The one deskilling result we cite does study sustained exposure [32], but no abstract in our corpus reports a dose-response curve, and, in practice, exposure is *ad libitum*. Only 530 claims specify outcome, population and horizon together, and, outside of students and clinical professionals, no cell holds more than four papers. One suggested remedy for deskilling in clinical settings is periodic reassessment. However, longitudinal measurement covered 18.3% of claims that state a horizon, dropping to 12.5% when the corpus is standardized on claim specificity.

This work is intended to inform an evidence-based approach to translating AI competencies into medical education and clinical practice. Six competency domains for clinician use of AI-based

tools have been defined by expert consensus, including evidence-based evaluation of such tools and practice-based learning and improvement [34], but they cannot be operationalized into staged autonomy or credentialing decisions without dose- and level-dependent evidence, which we show is largely absent from the abstract record. For decision makers and educators developing evidence-based guidelines, our results indicate that evidence on AI assistance ought to be read beyond abstracts when harms are at issue, and evidence tables should record whether each study measured and reported harm. Studies of AI assistance in training should name harm outcomes in advance and measure them over delayed and longitudinal horizons.

In future work, blinded human coding of 200 claims and a full-text check of 70 papers are planned. We also plan to operationalize the assistance-stage framework for which this audit was a prerequisite; this involves mapping degrees of machine involvement onto learning targets, training level and measurement horizon across medical education. The claim dataset and method can also track evidential form, a measurable property of a literature, over time in fast-moving areas such as clinical AI. Calls for longitudinal monitoring and adaptive curricula [33] presuppose some way of knowing whether harm measurement is keeping pace with deployment, and a claim-level audit is one such instrument. Until harm is specified as an outcome and measured over the interval across which learning is meant to persist, governance of AI assistance in clinical training risks continuing to interpret an absence of evidence as evidence of safety.

5 Limitations

All coding derives from abstracts, so every result here describes abstract reporting practice: the coverage gaps may be narrower in full text, and the horizon and statistical-marking results are lower bounds on what was actually measured.

Reliability is limited in four ways. Agreement figures are inter-model, not accuracy against expert judgment: reports that language models match or exceed expert coders [5] do not test whether high agreement between models reflects shared training data [6]. The `relation` field’s reliability under the final extraction prompt is unmeasured; its last measured value straddled the preregistered floor (κ 0.532 [0.433, 0.627]), so `relation` is excluded from clustering eligibility and the associative-framing result that depends on it is treated as provisional. The `valence` field that defines the exposure in every model has κ 0.631, just above the floor; random label error attenuates the estimates, but wording-dependent error, such as reading hedged statements as harmful, needs human coding to rule out. Finally, κ itself is a poor instrument for the sparse fields: near-chance κ can coexist with high observed agreement at extreme prevalence [19, 7], and κ presupposes predetermined, independent units — a condition claims segmented from abstracts by a model do not satisfy [8]. Sparse-field failures say as much about the statistic as about the fields.

Two analytic choices carry uncertainty. The latent class count is not identified by BIC; the substantive classes are stable in profile across neighboring solutions, but the exact partition should not be over-interpreted. The corpus has no reliable study-design variable: design is unavailable for 88% of papers, so adjustment and standardization use claim specificity, which describes the claim rather than the study. Recovering design labels from OpenAlex and PubMed would permit a stronger adjustment.

The corpus has 2 key limitations. The retraction screen is a lower bound — 99 claim-contributing sources carry no DOI and 9 DOIs did not resolve — and is re-run at submission. Retrieval is not saturated: a 500-record probe of the unscreened second citation generation yielded 12.8% includable records, and 943 screened records remain unresolved. Coverage results therefore describe the retrieved corpus; the benefit-harm asymmetry holds within both retrieval routes.

Data availability

The claim-level corpus and the derivation notebook are available under the MIT license, and the scripts that reproduce every table are in the supplementary material. Repository:

<https://github.com/vpatel143/Effect-of-AI-Assistance-on-Learning-Claim-level-evidence-audit>

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A Supplemental methods

A.1 Pre-window exception

Sources published before the review window are admitted only under a two-condition rule declared in configuration. Condition (i) is citation-based: the source must be cited by at least `min_citing_included_sources` (preregistered value 2) already-included in-window sources. Condition (ii) asks whether an in-window source establishes the same claim, and is a judgment about claim equivalence that remains with the author.

Condition (i) was decidable for 2,055 of 2,075 deferred records once citation linkage was available; reference lists were backfilled for 152 included records that lacked them, because a missing reference list undercounts citations and would wrongly reject a pre-window source. 704 records met condition (i).

A.2 Coding reliability

Per-field reliability was measured before the corpus-wide pass on a dual-extractor pilot (claude-haiku-4-5 against gpt-5.6-luna, a cross-family comparison, 156 claims stratified by role and claim specificity, seed 20260901), with 95% confidence intervals from a bootstrap that resamples source papers, as claims are correlated within a paper. Fields enter unsupervised analysis only if the interval clears $\kappa \geq 0.60$ and the field populates on at least 15% of claims. Three fields clear the floor with their whole interval: `object_type` 0.685 [0.603, 0.761] after its category glosses were replaced by an ordered rule, `population_role` 0.758 [0.662, 0.855], and `statistical_marking` 0.891 [0.763, 0.987]. Four sit above the floor, although their intervals straddle it: `direction` 0.688, `valence` 0.631, `population_setting` 0.616, `epistemic_strength` 0.683. `relation` falls below the floor 0.532, but the interval [0.433, 0.627] spans it. Two fail with intervals entirely below the floor: `subject_type` 0.446 and `subject_mode` 0.230. Four fields populate too rarely to cluster regardless of agreement: `mediator` (10.9%), `moderator` (10.1%), `magnitude_value` (10.3%) and `n_reported` (1.0%).

B Supplemental results

B.1 Exploratory claim analysis

Table A1: Descriptive statistics of claim schema across 6,063 claims. Levels are the number of distinct values in the field. “Modal” is the share of claims carrying the field’s single most frequent value. “Empty” is the share of claims where the field is `not_stated`, `not_applicable`, or blank.

Field	Levels	Modal %	Empty %
<code>subject_type</code>	11	55.8	18.1
<code>subject_mode</code>	8	50.2	24.6
<code>object_type</code>	10	28.4	15.0
<code>relation</code>	7	51.2	13.8
<code>direction</code>	6	57.4	19.3
<code>valence</code>	6	51.7	16.9
<code>population_role</code>	10	54.4	54.8
<code>population_setting</code>	8	54.9	55.4
<code>measurement_horizon</code>	6	82.1	84.1
<code>statistical_marking</code>	4	69.0	80.1
<code>epistemic_strength</code>	5	84.1	0.5
<code>object_entity</code>	4576	1.6	0.0
<code>mediator</code>	601	89.1	89.1
<code>moderator</code>	565	89.8	89.9
<code>magnitude_value</code>	458	89.7	89.7
<code>magnitude_unit</code>	146	86.9	89.8
<code>n_reported</code>	39	99.0	99.0
<code>evidence_span</code>	5921	0.1	0.0

B.2 Multiple correspondence analysis

Multiple Correspondence Analysis (MCA) over the 19 fields yielded 4 dimensions:

1. Dimension 1 (37.6%) — the 8 highest-contributing categories are all not stated / not applicable values, opposed to `relation=causes` / `valence=beneficial` / `direction=increase` / `subject_mode=use`. Driven by `direction` (16.1%), `valence` (15.6%), `relation` (15.4%).
2. Dimension 2 (13.7%) — null and non-significant findings with quantitative payload (`statistical_marking=not significant`, `relation=no relation`, `direction=no change`) against `relation=mediates`. `magnitude_value` and `magnitude_unit` contribute 12.7% each here.
3. Dimension 3 (12.7%) — `valence=neutral` + `direction=no change` + not significant against education-setting beneficial claims.
4. Dimension 4 (6.6%) — similar null-result contrast with `relation=moderates` at the other pole.

Field contributions confirm the same fields dominate throughout: `relation`, `direction`, `valence` and `statistical_marking` together carry 46–56% of every one of the first four axes. `object_entity` contributes $\leq 0.9\%$ to any axis, `mediator` 0.1% to `dim1`, `evidence_span` 0.0% to all of them.

After Benz cri adjustment the first dimension carries 37.6% of inertia. Its five highest-contributing categories are all sentinel values, and sentinels occupy eight of the top twelve, opposed to the modal substantive profile (`causes` / `increase` / `beneficial` / `use`). Dimensions 2 through 4 (13.7%, 12.7%, 6.6%) separate null and non-significant findings from positive ones.

B.3 Latent class analysis

We fitted latent class models with $K = 2\text{--}20$ over all 19 fields, taking the best of three expectation-maximization restarts at each K and using Bayesian Information Criteria (BIC) optimization. LCA revealed that the data supports approximately 9–12 classes, and we report $K = 11$ (entropy $R^2 = 0.93$).

Table A2: Substantive latent classes ($K = 11$ solution; four schema-artifact classes omitted). Percentages are within-class rates.

Class	n (%)	Profile
1	1162 (19.2)	Beneficial 96%, causes 89%, increase 90%, population unstated 82%
6	942 (15.5)	Education 93%, students 95%, beneficial 93%, increase 95%
0	704 (11.6)	Harmful 61%, decrease 50%, behavioral strategy 20%, associative 37%, hedged 18%, no statistical marking 91%
7	569 (9.4)	No intervention 85%, associative 59%, neutral 37%
3	434 (7.2)	Significant 66%, horizon stated 47%, magnitude unit present 51%, task performance 68%, beneficial 84%
9	338 (5.6)	Clinicians 53%, clinical or laboratory 79%, task performance 51%, harmful 18%
8	319 (5.3)	No relation 67%, no change 89%, not significant 49%, neutral 93%

B.4 Adjusted mutual information: top 10 pairs

Adjusted mutual information between field pairs, computed over all claims and over co-stated claims only.

1. `magnitude_value` $\sim$ `magnitude_unit` = 0.638
2. `direction` $\sim$ `valence` = 0.400
3. `population_role` $\sim$ `population_setting` = 0.332
4. `relation` $\sim$ `direction` = 0.245
5. `subject_type` $\sim$ `subject_mode` = 0.244

6. `relation ~ valence` = 0.198
7. `relation ~ statistical_marking` = 0.143
8. `relation ~ mediator` = 0.139
9. `subject_type ~ subject_entity` = 0.134
10. `relation ~ moderator` = 0.119

C Camera-ready sensitivity analyses

These analyses were added in revision and are produced by `eda/10_camera_ready.py` in the supplementary material. Data preparation and outcome definitions match Table 2, whose full-sample estimates the script reproduces.

Table A3: Table 2 restricted to generative-AI claims and to clinical claims (clinical setting or clinician population). Odds ratios [95% CI] for harmful versus beneficial claims; M0 pooled with paper-clustered errors, M3 within-paper conditional logit. Subset sizes are beneficial-or-harmful claims / papers. n/e = not estimable (fewer than five papers with within-paper variation).

Subset (claims / papers)	Outcome	M0	M3	M3 papers / claims
Generative AI (2,408 / 873)	Associative framing	3.00 [2.15, 4.20]	1.30 [0.80, 2.11]	66 / 267
	Hedged or speculative	2.69 [2.06, 3.51]	2.31 [1.57, 3.42]	108 / 433
	Significance-marked	0.18 [0.12, 0.27]	0.08 [0.03, 0.20]	49 / 213
	Carries a magnitude	0.41 [0.24, 0.71]	0.60 [0.26, 1.41]	22 / 90
Generative AI, 2023+ (2,329 / 836)	Associative framing	3.00 [2.13, 4.21]	1.26 [0.77, 2.06]	65 / 265
	Hedged or speculative	2.75 [2.10, 3.61]	2.57 [1.72, 3.84]	104 / 422
	Significance-marked	0.18 [0.11, 0.27]	0.08 [0.03, 0.20]	49 / 213
	Carries a magnitude	0.44 [0.25, 0.76]	0.60 [0.26, 1.41]	22 / 90
Clinical (352 / 193)	Associative framing	2.00 [0.74, 5.42]	n/e	4 / 11
	Hedged or speculative	3.11 [1.70, 5.68]	n/e	3 / 8
	Significance-marked	0.20 [0.09, 0.47]	n/e	4 / 11
	Carries a magnitude	0.57 [0.21, 1.49]	n/e	2 / 4
Generative AI × clinical (203 / 125)	Associative framing	1.74 [0.46, 6.52]	n/e	1 / 3
	Hedged or speculative	2.37 [0.95, 5.94]	n/e	0 / 0
	Significance-marked	0.14 [0.03, 0.59]	n/e	1 / 3
	Carries a magnitude	0.97 [0.32, 2.98]	n/e	0 / 0

Table A4: Papers that can contribute to the within-paper model (both a beneficial and a harmful claim) compared with papers that cannot. Claim specificity shares are means of within-paper shares.

	Single valence (excluded from M3)	Both valences (can contribute to M3)
papers	974	312
median year	2,025	2,025
% 2025-26	56.0	62.5
% direct search	17.7	25.0
% preprint	11.5	14.4
% generative AI	66.0	73.7
mean claims/paper	4.2	5.3
% taxonomic claims	28.5	30.7
% directional claims	60.7	63.8
% magnitude claims	10.7	5.5

Table A5: Added-noise test. A random share of valence labels (harmful $\leftrightarrow$ beneficial) is flipped and the models refitted; cells are median odds ratios over 20 replicates per rate, and the 0% column is the observed estimate. Random, non-differential coding error pulls every estimate toward 1.

Outcome	M0, share flipped				M3, share flipped			
	0%	5%	10%	20%	0%	5%	10%	20%
Associative framing	2.27	2.02	1.75	1.49	1.61	1.39	1.38	1.22
Hedged or speculative	2.44	2.08	1.92	1.52	2.02	1.71	1.60	1.35
Significance-marked	0.21	0.31	0.41	0.59	0.14	0.34	0.49	0.68
Carries a magnitude	0.43	0.52	0.61	0.71	0.67	0.71	0.87	0.83

Table A6: Table 2 by retrieval route: direct database search (channel 1) and inclusion-gated citation chasing (both passes). The harm share of beneficial-or-harmful claims is 25.9% for direct search, and 16.3% and 20.2% for the two chasing passes.

Route (claims / papers)	Outcome	M0	M3	M3 papers / claims
Direct search (789 / 250)	Associative framing	2.68 [1.59, 4.52]	2.10 [0.94, 4.68]	26 / 106
	Hedged or speculative	2.40 [1.62, 3.56]	1.70 [0.99, 2.93]	50 / 216
	Significance-marked	0.30 [0.17, 0.55]	0.27 [0.10, 0.72]	24 / 107
	Carries a magnitude	0.35 [0.17, 0.74]	0.53 [0.18, 1.57]	12 / 56
Citation chasing (3,090 / 1,036)	Associative framing	2.24 [1.67, 3.02]	1.49 [0.98, 2.27]	90 / 373
	Hedged or speculative	2.42 [1.89, 3.09]	2.17 [1.51, 3.11]	128 / 518
	Significance-marked	0.18 [0.12, 0.27]	0.09 [0.04, 0.23]	57 / 270
	Carries a magnitude	0.44 [0.24, 0.79]	0.75 [0.35, 1.62]	27 / 118

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